

Sports Performance Analysis Device

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Academic Year 2020–2021

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Abstract:

In this project, we designed and developed an intelligent device aimed at optimizing and improving the athletic performance of individuals. The device detects and analyzes the movement of each limb during physical exercises, allowing it to identify incorrect movement patterns and provide warnings and guidance on how to perform the exercises correctly. In this way, it can help prevent potential injuries caused by improper movements.

Furthermore, by analyzing the data stored in its memory, the device can evaluate users' performance improvements over a specified period and provide them with a performance report. This report includes information such as the number of sets completed, the number of errors, exercise execution speed, and changes in the range of motion of the user's limbs.

The device also accurately measures limb joint angles and calculates the range of motion of different joints. This capability could potentially allow the device to be used by orthopedic physicians as a tool for accurately assessing sports-related injuries.

1. Introduction:

If we look at the history of human societies and the development of human knowledge and scientific thought over the years, we can observe how the questions and challenges faced by people have evolved throughout history, along with their efforts to find appropriate answers. Problems have taken different forms in different periods, while the issues and concerns of each era have changed considerably. Some of these problems have even become almost unimaginable today.

However, there are also fundamental issues that have remained important regardless of time and place, with their underlying nature remaining largely unchanged. One such issue is the extent to which humans understand themselves. This forms the foundation of many fields related to human beings, including the humanities, education, mental health, and physical health.

One effective way to increase self-awareness is to record and analyze actions and movements over time. For this reason, many researchers have developed innovative devices and approaches for recording and analyzing physical condition, physical and mental health, individual abilities, and other aspects of human performance.

In this research, we propose a device and approach for more effective recording and analysis of individual performance in response to challenges associated with modern lifestyles. Although today's technology provides numerous undeniable benefits, it has also introduced new challenges. One of these is the emergence of increasingly stressful and busy lifestyles. [1,3] Combined with consumer culture, this lifestyle has contributed to reduced attention to physical health, resulting in unhealthy weight gain and decreased physical activity among members of society.

As this issue has become widespread both in our country and around the world and poses a serious threat to human health, researchers have sought ways to encourage people to adopt healthier lifestyles and increase their levels of physical activity. Existing studies have produced interesting findings, and various products have been introduced to the market for this purpose. These will be discussed in greater detail in the literature review section.

However, the effectiveness and practical benefits of currently available products have not yet been fully established, and further research is needed in this field.

2. Literature Review:

Advances in electrical and computer engineering, along with the development of various sensors for measuring changes in environmental and physiological parameters, have enabled the development and commercialization of a wide range of sports-related devices. These devices can monitor and analyze physical activities and provide users with more detailed information about their performance.

Such devices are commonly designed as smartwatches, fitness bands, or smart garments equipped with integrated sensing systems, making them convenient and portable for users. Some are also implemented as smartphone applications that collect data from sensors already available on mobile devices. After analyzing the collected data, these applications interpret the results in a user-friendly manner and present them to the user.

These devices can generally be divided into two main categories.

2-1. Daily Activity Tracking Devices

These devices monitor the user's daily activities and are primarily intended for individuals who do not engage in professional or specialized sports, with general health and physical activity being their main objectives. Two successful examples of such devices are discussed below.

2-1-1. Fitbit

Fitbit devices [2] are among the successful wearable technologies developed for monitoring people's daily activities. The device uses a three-axis accelerometer to detect human movement and count the number of steps taken. Using user-provided information, it can also estimate the distance traveled, calories burned, and number of floors climbed, while monitoring the duration and intensity of physical activities and displaying the results on its screen.

One of Fitbit's distinctive features is its ability to monitor sleep quality. By analyzing factors such as the duration of wrist inactivity and, in some cases, breathing rate, the device can estimate parameters such as the time required to fall asleep and the amount of time spent in deep sleep.

Fitbit also provides dedicated applications for smartphones and computers. When the device is synchronized with these applications, the collected data can be stored in cloud storage. The platform also provides an online network for users, allowing them to compare their activity and performance with those of other users.



Figure 2-2. Fitbit Application



Figure 2-1. Fitbit Wristband

2-1-1. Moves App

Moves [4] is a smartphone application designed to track users' daily activities, including distance traveled, activity duration, number of steps, and calories burned during three types of activities: running, walking, and cycling.

The application was designed to encourage users to increase their physical activity. Its main goal is to encourage users to reach 10,000 steps per day and thereby develop long-term healthy habits.

Another feature of the application is its ability to record the places users visit throughout the day and the amount of time they spend at each location. It also tracks how users travel between different locations.

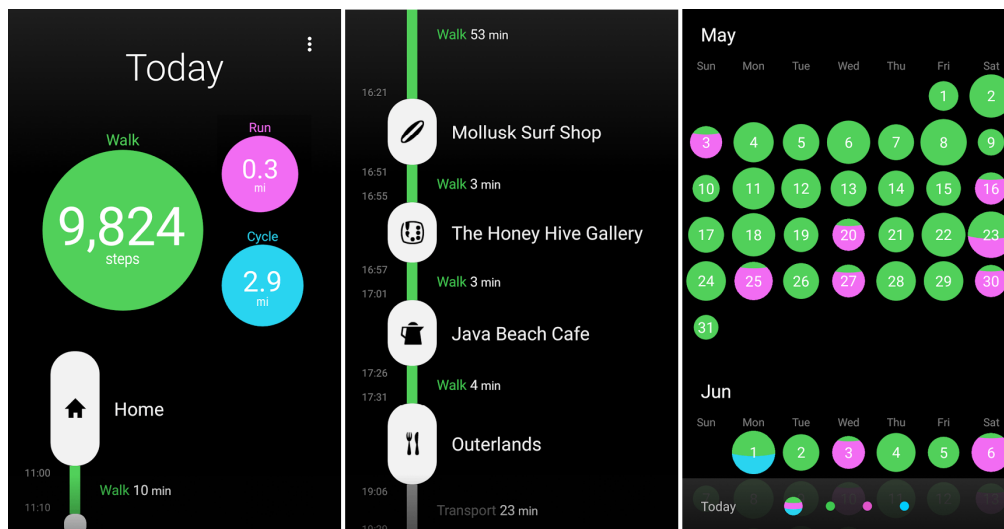


Figure 2-3. Moves Application Interface

2-2. Sports Performance Analysis Devices

Unlike daily activity tracking devices, which are primarily goal-oriented, sports performance analysis devices focus on the quality of athletic performance and methods for improving it. They help athletes improve their performance over a specific period of time.

These devices are also available in more specialized forms for professional sports such as running, cycling, surfing, and others. Accordingly, they can monitor additional parameters such as body weight, Body Mass Index (BMI), exercise intensity, and movement speed. Some examples are discussed below.

2-2-1. Garmin

Different Garmin devices [5] are equipped with various sensors depending on their intended use, including GPS, pedometers, barometers, and accelerometers. They can measure parameters such as calories burned, distance traveled, and heart rate.

These devices support various sports, including running, swimming, and cycling. When connected to a computer, they can display route maps and maintain an archive of previously traveled routes. The devices can be connected to computers via Bluetooth, USB, or Wi-Fi.



Figure 2-5. Garmin Fitness Band Application



Figure 2-4. Garmin Fitness Band

2-2-2. Polar:

Polar devices [6] provide different programs tailored to various types of sports. Based on the data they collect, they offer athletes recommendations and guidance for improving their athletic performance.

These devices can also serve as rehabilitation assistants and can be used for purposes such as weight management and maintaining a healthy lifestyle.

Another distinguishing feature of these devices is the availability of various programs, including endurance training guidance, fitness tests, smart coaching, and relaxation tests. These features enable a more detailed assessment of an individual's physical condition.

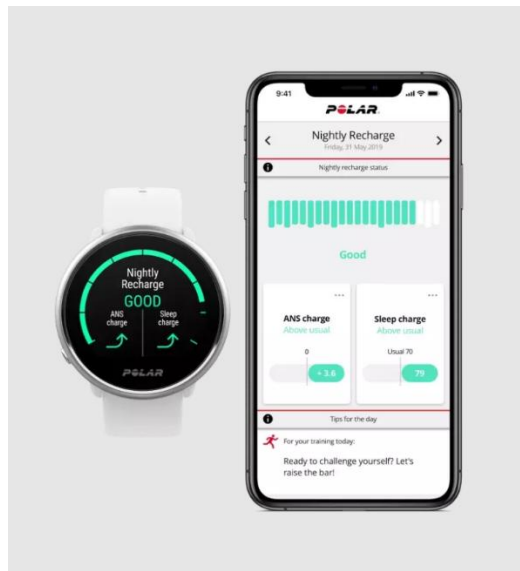


Figure 2-7. Polar Fitness Band and Dedicated Application



Figure 2-6. Polar Fitness Band

2-3. Applications and Psychological Effects of Existing Devices

In addition to research focused on the development of smart devices, studies have also investigated their effectiveness and psychological effects on users. However, it should be noted that research in this area has so far been relatively limited, and further investigation is required to develop more reliable solutions to address the shortcomings of these devices.

Studies have shown that approximately one-fifth of smartphone users use sports-analysis applications and devices, with the average period of use being approximately six months [7]. After this period, usage tends to decline for various reasons, including improper device functionality, inaccurate or unreliable data and analyses, the lack of mechanisms for maintaining user motivation throughout the period of use, and failure to adequately meet users' needs in achieving their desired results [8].

Athletes and individuals who participate in sports out of personal interest tend to use these devices more frequently than other users. This group also generally reports greater satisfaction with their effectiveness and their contribution to improving athletic performance. In addition, people who have previously used such devices tend to show greater willingness to continue using them compared with those encountering this type of technology for the first time [3].

As mentioned above, the use of these devices can generally be divided into two areas: daily activities and specialized sports. Individuals who use these devices for everyday activity rather than specialized sports are generally less motivated and have less specialized knowledge in sports. Therefore, incorporating mechanisms for maintaining motivation and providing users with information about how to improve their performance is essential for encouraging continued use.

Various approaches have been proposed in this area, one of the most successful being Goal-Setting Theory [9]. This theory examines how having specific goals can influence individuals' progress and the rate at which they improve, how goals should be established according to individual differences, and how different people respond to goal-setting.

Based on this theory, many existing devices use goal-oriented structures in which users are assigned goals of different scales and difficulty levels in a step-by-step manner, ultimately leading toward a long-term objective. These goals should be sufficiently challenging to engage users and encourage them to work toward achieving them. At the same time, they should not appear impossible or unattainable. Individual preferences and circumstances are also important factors that should be considered when determining the type of goal and the appropriate strategy for achieving it.

This issue takes on a different form in professional sports. According to existing research, athletes use sports-analysis devices for longer periods, and current devices have been able to meet some of their needs [3]. However, one of their main limitations is the relatively narrow range of sports they can analyze. Their applications are generally limited to activities such as walking, running, swimming, cycling, and skiing, making them less useful for other sports [3].

Furthermore, these devices generally cannot provide detailed, sport-specific analysis of individual movements. Instead, they mainly provide general information such as total exercise duration, average speed, and distance traveled. Given the importance of performance quality in achieving athletic goals, such information is insufficient for developing an optimized training program tailored to an individual athlete.

Another important concern in specialized sports is preventing injuries during intensive training. Such injuries can occur because of improper exercise techniques, carelessness, or insufficient knowledge of correct movement patterns. They can interrupt an athlete's progress and, in some cases, pose a serious risk to their health.

Therefore, an effective warning mechanism that alerts users when exercises are being performed incorrectly could play an important role in fulfilling the requirements of a comprehensive sports-analysis device. Such a feature is generally absent from existing devices.

3. Problem Statement

Based on the research conducted, it can be concluded that existing sports-analysis devices have not been sufficiently designed with long-term use in mind and cannot yet be considered comprehensive assistants for improving athletic performance.

Although these devices each provide specific features and have contributed to improvements in the analysis and evaluation of athletic performance, they still do not fully address the needs of either specialized or non-specialized users. They do not provide sufficient information about individual performance to allow users to independently monitor the correctness and quality of their movements without relying on a coach or instructor.

In addition, users are often not provided with sufficient analysis to identify their athletic level, abilities, or potential, making it difficult to develop a personalized long-term training program.

Another challenge faced by athletes is sports-related injury. Our investigation showed that a significant number of sports injuries can result from performing exercises incorrectly. Some examples are discussed below.

3-1. Sports Injuries

Sports injuries can generally be divided into two categories: acute injuries and chronic injuries.

Acute injuries occur suddenly and typically involve a high level of force or impact. Examples include bone fractures, tendon or muscle strains, ligament sprains, and contusions. These injuries are usually accompanied by pain and may temporarily prevent the affected body part from functioning normally.

The second category consists of chronic injuries, which develop gradually over a longer period, often as a result of excessive or improper use of body parts. Examples include tendinitis, bursitis, stress fractures, and incomplete fractures. Like acute injuries, these conditions may be accompanied by pain, swelling, and impaired function of the affected area [10].

This research focuses primarily on chronic injuries and on preventing injuries that develop over time as a result of improper movement patterns. The following sections describe several examples of injuries that the proposed device is intended to help prevent.

3-1-1. Rotator Cuff Injury

The rotator cuff [1] is a group of muscles and tendons surrounding the shoulder joint that provides stability to the shoulder. Injuries to this tissue are common in sports involving strenuous or repetitive overhead arm movements, such as volleyball, baseball, swimming, tennis, and weightlifting.

In these sports, rapid and forceful shoulder movements can cause rotator cuff tears; however, repetitive and strenuous movements are generally a more common cause. Recovery typically takes at least 3 to 6 months. In severe cases requiring surgery, recovery may take approximately 18 months, and complete recovery is not always guaranteed [11, 12].

3-1-2. Shoulder Joint Pain

Shoulder pain is common in sports that involve repetitive shoulder movements, such as tennis and baseball. It may result from shoulder joint inflammation or frozen shoulder [2]. It may also be caused by shoulder impingement syndrome [3], which can occur in sports such as swimming and tennis.

Recovery time varies depending on the underlying cause. Frozen shoulder may take up to 18 months to recover, while recovery from shoulder impingement syndrome typically takes around 3 to 6 months. In cases of shoulder inflammation, recovery may take up to 6 months [11, 13].

3-1-3. Elbow Bursitis

Elbow bursitis can result from a direct impact to the elbow or from repetitive elbow movements. It is common in sports such as tennis, swimming, and baseball. Treatment may take anywhere from several days to several months, and in cases requiring surgery, recovery may take up to 6 months [14].

3-1-4. Elbow Tendon Injury

Excessive strain on the muscles on the outer side of the forearm, particularly when the wrist is extended backward to resist an external force, can lead to elbow injuries. Such injuries are often associated with repetitive elbow movements, including hitting a ball in golf, weightlifting, CrossFit, and tennis.

Recovery time varies from several weeks to several months. In cases requiring surgery, recovery may take approximately 4 to 6 months [11].

3-1-5. Wrist and Hand Disorders

The wrist is a complex joint with a wide range of motion. However, its high mobility and relatively low stability also make it particularly vulnerable to injury. Injuries to the tendons of the hand and wrist are common in many sports, with some resulting from repetitive use and excessive stress on the wrist.

Three common conditions are De Quervain's tenosynovitis [4], intersection syndrome [5], and carpal tunnel syndrome [6]. These conditions are frequently associated with sports such as tennis, skiing, weightlifting, and bodybuilding. Recovery from these injuries may take at least six months [13].

The injuries discussed above highlight the importance of prevention and proper instruction on how to avoid them. Therefore, one of the primary objectives of the proposed device is to help prevent such disorders in specialized sports.

However, as mentioned earlier, prevention is primarily possible for injuries that develop gradually over time. Sudden injuries are generally beyond the system's preventive capabilities because of their

unpredictable nature. Nevertheless, this type of technology can also be useful in the examination and assessment of such injuries, as discussed in the following section.

3-2. Physical Assessment and Diagnosis of Sports Injuries

One of the examinations performed on an injured body part to determine the type and severity of an injury is the assessment of its range of motion (ROM). The assessment procedure varies depending on the body part being examined.

For example, shoulder range of motion can be evaluated in two positions: with the arm raised vertically overhead and with the arm resting downward. The assessment includes measuring the maximum range of motion, evaluating the movement pattern of the arm, and determining the level of pain experienced when moving beyond the normal range of motion.

The angle between the upper arm and the torso can be used to assess shoulder range of motion. The range of internal and external rotation of the arm is also important. Moving the arm around the shoulder can provide valuable information about shoulder involvement and stability.

Pain occurring when the arm is rotated through approximately 80° to 120° relative to the shoulder can be an indicator of tendon inflammation associated with impingement. This range is commonly referred to as the painful arc. In addition, the angle of arm movement relative to the shoulder can be used to assess muscle strength, which is one of the parameters considered when evaluating rotator cuff injuries [15].

Elbow range of motion and strength are also evaluated during physical examinations. For this assessment, the arm is placed in a resting position with the elbow extended, and the angle between the upper arm and forearm is measured. This angle is estimated to be approximately 10° in men and 13° in women on average.

The movement of the forearm is an important indicator of elbow function and can be evaluated along two primary axes: elbow flexion and extension, and forearm rotation inward and outward. The average range of elbow flexion and extension is approximately 0° to 140°. Forearm pronation is approximately 70°, while supination ranges from approximately 85° to 90°. Supination strength is also approximately 15% greater than pronation strength [15].

Similar assessments can be performed for other joints, and the resulting measurements can be used to assist in identifying and evaluating injuries.

As mentioned previously, the proposed device can use range-of-motion data to support the examination and assessment of musculoskeletal conditions. It could allow physicians to perform relevant measurements more conveniently than traditional methods, such as using a goniometer or inclinometer [16].

Following the analysis of these injuries and the preparation of a list of sports movements that may cause long-term injuries when performed repeatedly, the development of the device and its accompanying application was initiated. The following sections describe the development process in detail.

4. Main Problem-Solving Strategy:

Considering the limitations discussed in the previous section regarding the effectiveness and responsiveness of existing devices designed to analyze human performance in both sports and non-sports contexts, we decided to develop a device that could address some of the shortcomings of this emerging technology. By focusing more closely on the details of athletic movements, the proposed system was intended to provide more accurate assessments and more reliable results.

Since injuries resulting from strenuous physical activities can have serious consequences and pose risks to individuals' health, we conducted research and consulted healthcare professionals specializing in physiotherapy and sports medicine. We asked for their opinions regarding the potential effectiveness of such a system in preventing sports-related injuries. After receiving positive feedback regarding the potential practical value of the proposed system, we proceeded to the implementation stage.

First, sports-related injuries were investigated, and their causes and potential prevention methods were identified. Based on these findings, a list of relevant sports and exercises was compiled. For selecting the exercises to be included in the application, two main categories were considered. The first category consisted of exercises that can be easily performed at home without specialized exercise equipment. These exercises were intended for individuals who are new to physical activity or do not pursue sports professionally, allowing them to independently monitor their movements and receive guidance on performing them correctly. The second category consisted of specialized exercises performed using sports equipment, where incorrect use or execution may result in injury.

4-1. Application Design

In this section, we developed a sports-oriented mobile application. We first became familiar with Java programming and the principles of designing and developing Android applications using Android Studio [1], learning the concepts and techniques required to meet the needs of the project. We then designed the application's functional structure and proceeded with its implementation.

The application consists of two main components. The first is the user interface, which includes the information and features with which the user directly interacts. The visual appearance and interface elements of the application are designed in this component, and the analyses performed by the device are ultimately presented to the user through the interface.

The second component consists of the processes that take place in the background and are not directly visible to the user. The details of the functionality implemented in each component are described below.

The user interface consists of three main screens:

1. Home
2. Exercise List
3. Exercise Details

On the Home screen, there is a button labeled Activities. When the user selects this button, the application opens the exercise list and simultaneously attempts to establish a Bluetooth connection with the device. Once the Bluetooth connection is successfully established, a Toast message [2] stating

"Bluetooth connected" is displayed. If the connection cannot be established, the message "Bluetooth connection failed. Please try again." is displayed.

On the Exercise List screen, the available exercises are displayed as a series of cards. Each card contains the name of the exercise and a GIF demonstrating the correct execution of the movement. By selecting a card, the user is taken to the dedicated page for that exercise, where detailed instructions are provided.

At the top of the exercise page, a GIF demonstrates the correct execution of the movement. Below it, instructions for using the system and attaching the device to the relevant body part are provided. After attaching the device to the target body part, the user can press the Start button and begin performing the exercise. The exercise can be stopped at any time by pressing the Stop button.

During the exercise, the number of sets completed, the measured range of motion, and the movement speed of the relevant limb are displayed simultaneously in the application. If the measured range of motion deviates from the acceptable range, the corresponding value is displayed in red. A description of the detected error and instructions for correcting the movement are also displayed in red at the bottom of the screen, allowing the user to understand what went wrong.

In addition to the information provided during the exercise, once the user stops the exercise, the application displays the user's performance progress as a percentage, allowing them to compare their current performance with previous sessions and determine whether their performance has improved or declined.



Figure 4-4. Medical Assessment Screen



Figure 4-3. Exercise Details



Figure 4-2. Exercise List



Figure 4-1. Home Screen

4-2 Application Dependencies and Interface Implementation

The following dependencies [1] were used in the design of the application's user interface to incorporate GIFs and implement the required application components, including components provided by Google [17]:

```
1. implementation 'pl.droidsonroids.gif:android-gif-drawable:1.2.19'  
2. implementation 'com.google.android.material:material:1.3.0'
```

The calculation of the range of motion, set count, error detection, and Bluetooth communication is performed simultaneously in the background of the application.

To receive data through the smartphone's Bluetooth interface, the application must be granted permission to access Bluetooth. The following permissions were therefore included:

```
1. <uses-permission  
2.     android:name="android.permission.BLUETOOTH" />  
3. <uses-permission  
4.     android:name="android.permission.BLUETOOTH_ADMIN" />
```

The following classes, which provide the functionality required for Bluetooth communication, were used to establish a connection between the smartphone and the device:

```
1. import android.bluetooth.BluetoothAdapter;  
2. import android.bluetooth.BluetoothDevice;  
3. import android.bluetooth.BluetoothSocket;
```

When the Activities button is selected, the application uses the functionality provided by these classes to detect the available Bluetooth devices in its surroundings. After the MAC address of the target device is selected, the application establishes a communication channel with the device for data transmission.

Communication between the two Bluetooth devices is bidirectional, meaning that either device can write data to or read data from the communication channel, byte by byte.

The application receives data from the device via Bluetooth in the form of character-based packets. The received data are stored in an array and subsequently converted into integer values. The parameters received by the application consist, in order, of yaw, pitch, and roll, representing angular changes about the three axes, followed by the angular velocity about each of these axes.

After the data are received, the application calculates the range of motion. Based on the measured range of motion and the specific execution pattern of each exercise, the number of completed sets is determined. If the measured range of motion falls outside the acceptable range, the application detects an error. In addition to changing the corresponding range-of-motion value to red and displaying details of the detected error to the user, the application sends the error signal back to the device through the Bluetooth communication channel, causing the warning buzzer to sound.

The method used to calculate the number of sets and detect errors varies depending on the specific exercise.

4-2-1 Data Filtering and Hysteresis

To prevent minor fluctuations in the data received from the device from affecting the results, the 100 most recent values of each parameter are stored in an array, and their average is considered the valid measurement. This makes the processed data considerably more reliable.

This method of reducing rapid fluctuations acts as a low-pass filter [2], specifically a moving average [3].

In addition, to prevent small fluctuations around the upper and lower limits of the acceptable range from generating repeated and annoying warning sounds, two different thresholds are defined for each parameter: one threshold for entering the invalid region and another for returning to the valid region.

For example, if an angle exceeds 100°, the device issues a warning. Once the warning has been triggered, however, the value must fall below 90° before the warning is cleared. In other words, hysteresis [4] is introduced for this parameter.

4-2-2 Error Detection

Error detection in the application is based on the standard range-of-motion values presented in the following table [16, 18]. In addition, experienced sports coaches were consulted, and their expertise and practical experience in performing the relevant exercises were incorporated into the error-detection process.

0-125	Flexion	Hip	1
0-115	Extension		
0-45	Abduction		
0-135	Flexion	Knee	2
0-120	Extension		
0-25	Rotation	Foot	3
0-160	Flexion		
0-145	Extension	Elbow	4
0-90	Internal Rotation		
0-90	External Rotation		
0-50	Extension	Shoulder	5
0-90	Flexion		
0-90	Abduction		
0-90	Transverse Rotation		
0-90	Medial Rotation		
0-90	Flexion	Wrist	6
0-70	Extension		
0-25	Abduction		
0-65	Adduction		

Table 1. Range-of-Motion Variations of Different Body Parts

To perform the operations described above in the user interface and the background section of the application simultaneously, separate threads [1] were used. This allows the two parts of the code to execute approximately concurrently.

4-2. Electronic Circuit of the Device

The electronic circuit of the device consists of an Arduino Nano board [2] as the processing core, an MPU6050 module [3] for detecting limb movements, an HC-05 Bluetooth module for wireless communication, and a buzzer [4] for providing audible warnings to the user.

One of the reasons for selecting these components was their relatively low cost compared with other available modules. In addition, their small size and low weight allow the device to remain compact and lightweight. This makes it easier to attach the device to the user's limb and carry it during physical activity.

The following sections describe each of these components in detail.

4-2-1. MPU6050 Module

An MPU6050 module is used to detect limb movements. The module contains an MPU-6050 integrated circuit (IC) [5] along with the peripheral circuitry required for its operation. The IC includes a three-axis accelerometer and a three-axis gyroscope [6].

By combining information from these two sensors, the IC determines angular displacement around the x, y, and z axes and transmits the corresponding data to the device's microcontroller, where the movement of the limb is analyzed.

To obtain angular displacement and angular velocity along the three coordinate axes, the functions provided by the MPU6050_6Axis_MotionApps20.h library were used.

Communication between the MPU6050 and the microcontroller is implemented through the I²C serial communication protocol. This provides bidirectional but asynchronous communication. The application uses the I2Cdev.h and Wire.h libraries to initialize and operate this communication interface [19].

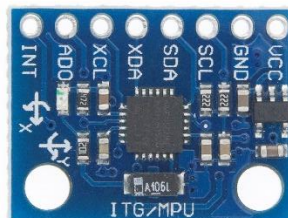


Figure 4-4. MPU6050 Module

4-2-2. Bluetooth

To establish wireless communication between the electronic circuit and the application, an HC-05 Bluetooth module was used. The HC-05 is a Class 2 Bluetooth device and provides a communication range of approximately 10–20 meters, which is sufficient for the requirements of this device.

The data transmission rate of these Bluetooth modules can reach up to approximately 1 Mbps, providing sufficient communication speed for the system. The Bluetooth module connects to the

smartphone and transmits hardware data to the application. When necessary, it can also receive warning commands from the application.

Communication between the Bluetooth module and the microcontroller is implemented through a serial UART connection. The SoftwareSerial.h library was used to establish and operate this communication interface.

The reason for using this library and configuring Arduino Nano pins as software-based Tx and Rx pins was that the primary TX and RX pins of the Arduino Nano were being used for serial communication between the microcontroller and the computer during testing and development. Using these pins for Bluetooth communication could therefore interfere with data transmission between the two Bluetooth devices [20].

Since the smartphone's Bluetooth is configured as the master by default, the device's Bluetooth module must be configured as the slave. Therefore, the HC-05 module must be configured to operate in Role 0. This configuration can be performed while the Bluetooth module is in AT mode, using the commands specific to the HC-05.

The data transmission process is as follows: the microcontroller first receives data from the MPU6050 and transmits the data through its Tx pin to the Bluetooth module using the UART protocol. The Bluetooth module then wirelessly transmits the data to the Bluetooth module on the receiving device, which receives the data and makes it available to the application.



Figure 4-5. HC-05 Bluetooth Module

4-2-3. Arduino Nano

An Arduino Nano board is used to control the operation of the device. The Arduino Nano is a smaller and simpler version of the Arduino Uno board [14]. Due to its compact dimensions, it is well suited for this application.

Since the computationally intensive tasks of the system are performed within the application and by the smartphone's processor, the capabilities provided by the Arduino Nano are sufficient, and a more powerful processing unit is not required.

Another advantage is Arduino's user-friendly development environment, which simplifies working with microcontrollers. The board is programmed using the company's Integrated Development Environment (IDE) [7], known as the Arduino IDE, which supports the C++ programming language [21].

The Arduino Nano contains an ATmega328 microcontroller. The role of the microcontroller in the circuit is to perform the required operations and control their timing. It receives data from the MPU6050 and performs the necessary calculations to determine the angular displacement and angular velocity around the three coordinate axes. The resulting data are then transmitted to the application via Bluetooth.

When the microcontroller receives a command from the Bluetooth module indicating that the warning buzzer should be activated, it sets the voltage level of the pin connected to the buzzer to HIGH, causing the buzzer to sound.

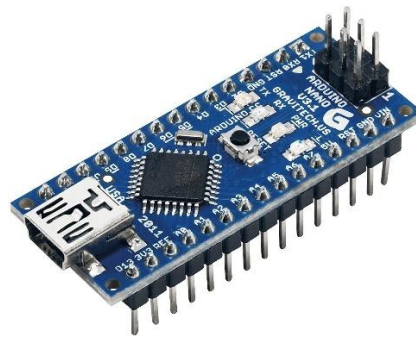


Figure 4-6. Arduino Nano Board

4-2-4. Battery

A 9 V battery is used to power the electronic circuit. The voltage produced by the battery is higher than the 5 V supply voltage required by the Arduino Nano. The onboard voltage regulator [8] reduces the input voltage to the required 5 V level.

The 5 V output supplied by the Arduino Nano powers both the Arduino board itself and the Bluetooth and MPU6050 modules.



Figure 4-7. 9 V Battery

4-2-5. Warning Buzzer

The buzzer is used to immediately notify the user when the system detects an error in the execution of an exercise. Once an incorrect movement is detected, the buzzer sounds, allowing the user to recognize the error quickly and correct the movement.

Because the buzzer produces a relatively loud sound, it is also suitable for providing warnings in noisy environments, such as gyms and sports facilities.



Figure 4-8. Warning Buzzer

4-2-6. Printed Circuit Board (PCB)

The electronic circuit of the device was designed using Altium Designer [10]. Its intelligent design features and ease of use made it a suitable choice compared with other available PCB design software.

An effort was made to design the printed circuit board (PCB) [9] as a single-layer board in order to reduce the time and cost required for manufacturing.

After completing the PCB design, the generated manufacturing files were sent to a PCB manufacturer. Once the board was fabricated, the required components were manually soldered onto the board.

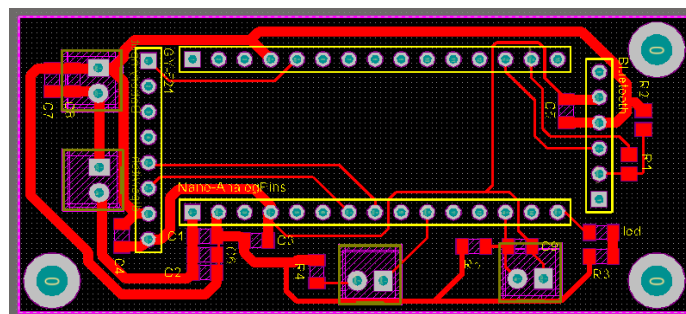


Figure 4-9. PCB Designed for the Electronic Circuit

4-2-7. Device Enclosure

The device enclosure is made of plastic, which was selected because of its low weight while still providing adequate mechanical strength.

The enclosure was purchased as a ready-made housing. A power ON/OFF switch was installed on the exterior of the enclosure so that the device could be turned on or off without opening the housing.

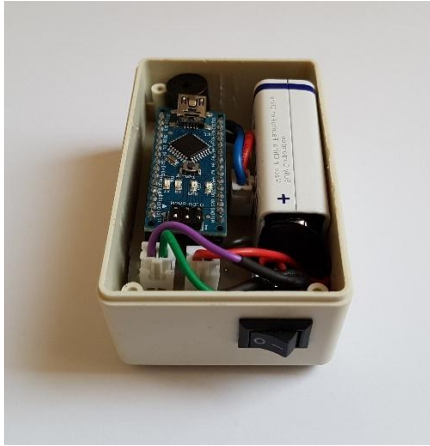


Figure 4-11. Internal View of the Device



Figure 4-10. External View of the Device

5. Applications and Results

The proposed device can be useful for several groups of users. The first group consists of beginners who intend to engage in physical activity for non-specialized purposes. The application includes exercises that can be performed at home without specialized sports equipment, allowing users to engage in light daily exercises while independently monitoring the correctness of their movements.

The second group consists of individuals who wish to engage in sports activities more professionally and with the assistance of exercise equipment, but who do not have sufficient knowledge or experience in the relevant sports. Such individuals may require an experienced coach to make meaningful progress; however, they may choose not to work under the supervision of a coach because of limited access to qualified coaches or the high cost of personal training. For these users, the proposed device serves as an educational guide. Initially, the application introduces the user to the correct execution of each movement through GIFs and videos. During exercise, the system then provides warnings whenever an incorrect movement is detected, helping the user gradually learn and maintain the correct movement pattern.

Another group that can benefit from this device is professional athletes. Because of the intensive training they undergo, professional athletes may be exposed to a higher risk of sports-related injuries. By using the designed warning system to detect movements that fall outside the defined safe range, the device may help users identify potentially problematic movements before they result in more serious injuries.

In addition to the applications described above, the capabilities of this device can also be used in the examination and assessment of sports-related injuries. These capabilities provide a relatively simple means of evaluating movement patterns, changes in range of motion, and movement strength, and may be useful to orthopedic specialists and other physicians.

5-1. Main Results of the Project

As discussed earlier, the emergence of smart devices and wearable technologies in areas such as healthcare, sports, and everyday activities has made the monitoring and analysis of physical processes considerably easier and more precise. However, these technologies still have considerable potential for further development before they can fully address the diverse needs of users.

For this reason, we developed a device with a different set of capabilities compared with existing solutions. The main features of the proposed system are as follows:

Error detection during exercise: The system detects incorrect movements and informs the user, providing additional information such as the number of errors during an exercise session and the type of error detected.

Range-of-motion measurement: The system measures changes in the range of motion along three different axes.

Movement-speed measurement: The system measures the speed of movement of the target limb.

Performance monitoring: The system calculates the user's improvement or decline relative to previous exercise sessions.

Set counting: The system counts the number of completed sets of an exercise.

Movement instruction: The application provides guidance on the correct execution of exercises.

The proposed sports device can serve as a guide and assistant for different types of users depending on their objectives and can improve the process of monitoring and evaluating exercise performance.

5-2. Future Research

In future work, the proposed system could be expanded to cover a wider range of sports and exercises by increasing the number of exercises available in the application. Furthermore, the data collected by the device could be used to perform more sophisticated analyses.

For example, by analyzing data collected over an extended period, the system could estimate the user's level of athletic performance and personalize exercises according to the individual's performance level. If the user specifies a particular goal, the system could monitor their progress toward that goal and recommend a training program tailored to the individual's capabilities and objectives. It could also provide recommendations for improving training efficiency and achieving better results.

Another potential capability of the device is the measurement of the actual duration of exercise. This measurement could be more accurate than simply recording the total session duration because the system would count time only when it detects that an exercise movement is actually being performed. This would allow users to determine the amount of time they have effectively spent exercising and evaluate their performance more accurately.

Using a smartphone as the user interface for the device also provides the possibility of connecting the system to the Internet. This could allow users to access up-to-date information relevant to physical activity, obtain broader statistics and analyses regarding exercise and its effects, and establish a social network connecting coaches, athletes, and beginners.

Such a platform could help users gain a better understanding of their progress and training path while allowing them to benefit from the experience and expertise of qualified coaches and professional athletes.

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